

Syntax and Truth Table of Propositional Logic

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- 1 Syntax
- 2 Truth Function and Table

In Propositional Logic:

- We express arguments in terms of propositions.
- So, we need a language to write propositions.
- Such the language is called "Propositional Language"

In Propositional Logic:

- We express arguments in terms of propositions.
- So, we need a language to write propositions.
- Such the language is called "Propositional Language"

Fact

Language = Syntax + Semantics

Propositional Language: simple statements

Rule:

Use upper-case letters to represent **simple statements**

Examples

B: Butane is a hydrogen compound.

J: James Joyce wrote Ulysses.

P: Parakeets are colorful birds.

T: The monk seal is threatened with extinction.

Propositional Language: compound statements

Rule:

To represent **compound statement**: use operators (or connectives) to relate simple statements.

Examples

M: Mai has programming skill.

H: Hoa has programming skill.

$M \wedge H$: Mai **and** Hoa have programming skill.

Propositional Language: compound statements

Rule:

To represent **compound statement**: use operators (or connectives) to relate simple statements.

Examples

M: Mai has programming skill.

H: Hoa has programming skill.

$M \vee H$: **Either** Mai **or** Hoa has programming skill.

Propositional Language: compound statements

Rule:

To represent **compound statement**: use operators (or connectives) to relate simple statements.

Examples

M : Mai has programming skill.

H : Hoa has programming skill.

$\neg M$: Mai does **not** have programming skill.

$\neg H$: Hoa does **not** have programming skill.

Propositional Language: compound statements

Rule:

To represent **compound statement**: use operators (or connectives) to relate simple statements.

Examples

M: Mai has programming skill.

H: Hoa has programming skill.

$M \Rightarrow H$: If Mai has programming skill **then** Hoa has programming skill.

Propositional Language: compound statements

Rule:

To represent **compound statement**: use operators (or connectives) to relate simple statements.

Examples

M: Mai has programming skill.

H: Hoa has programming skill.

$M \Leftrightarrow H$: Mai has programming skill **if and only if** Hoa has programming skill.

Propositional Language: compound statements

Rule:

To represent **compound statement**: use operators (or connectives) to relate simple statements.

Operators or Connectives

$\neg$: Negation

$\wedge$: Conjunction

$\vee$: Disjunction

$\Rightarrow$: Implication

$\Leftrightarrow$: Equivalence

Propositional Language: Operators

Operator	Name	Logical Function	Used to translate
$\neg$	neg	negation	not, it is not the case that
$\wedge$	wedge	conjunction	and, also, moreover
$\vee$	vee	disjunction	or, unless
$\Rightarrow$	rightarrow	implication	if... then... , only if
$\Leftrightarrow$	leftrightarrow	equivalence	if and only if

Propositional Language: Exercises

Questions and Exercises here!

Propositional Language: Formal definition

Definition

A Propositional Language L is set of L -formulas. L -formulas are generated by using the following rules:

- 1 All of simple statements are L -formula.
- 2 If A is an L -formula, then $(\neg A)$ is an L -formula.
- 3 If A and B are L -formulas, then $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, and $(A \leftrightarrow B)$ are L -formulas.

Truth Table

p	$\neg q$
F	<u>?</u>
T	<u>?</u>

Truth Table

Operator: Negation

p	$\neg p$
F	T
T	F

Truth Table

Operator: Conjunction

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

Truth Table

Operator: Disjunction

p	q	$p \vee q$
F	F	F
F	T	T
T	F	T
T	T	T

Truth Table

Operator: Implication

p	q	$p \Rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Truth Table

Operator: Equivalence

p	q	$p \Leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

Truth Table

for compound statement

How do we evaluate their value?

Example: $(A \Rightarrow B) \wedge C$

Truth Table

$s \triangleq$

A	B	C	s
F	F	F	$\underline{\underline{?}}$
F	F	T	$\underline{\underline{?}}$
F	T	F	$\underline{\underline{?}}$
F	T	T	$\underline{\underline{?}}$
T	F	F	$\underline{\underline{?}}$
T	F	T	$\underline{\underline{?}}$
T	T	F	$\underline{\underline{?}}$
T	T	T	$\underline{\underline{?}}$

Truth Table

$$s \triangleq (A \Rightarrow B) \wedge C$$

A	B	C	$(A \Rightarrow B)$	s
F	F	F	T	$\frac{?}{=}$
F	F	T	T	$\frac{?}{=}$
F	T	F	T	$\frac{?}{=}$
F	T	T	T	$\frac{?}{=}$
T	F	F	F	$\frac{?}{=}$
T	F	T	F	$\frac{?}{=}$
T	T	F	T	$\frac{?}{=}$
T	T	T	T	$\frac{?}{=}$

Truth Table

$$s \triangleq (A \Rightarrow B) \wedge C$$

A	B	C	$(A \Rightarrow B)$	s
F	F	F	T	F
F	F	T	T	T
F	T	F	T	F
F	T	T	T	T
T	F	F	F	F
T	F	T	F	F
T	T	F	T	F
T	T	T	T	T

Truth Table

Another statement

$$s \triangleq (A \Rightarrow B) \wedge A$$

$$s \triangleq (A \Rightarrow B) \wedge A$$

A	B	$(A \Rightarrow B)$	s
F	F	T	F
F	T	T	F
T	F	F	F
T	T	T	T